

TO: Consulting and Analytics Club, IIT GUWAHATI
FROM: Krishna Vijay Kunwar
RE: Top Bottleneck Hubs, FTL/Carting Routing Correction, and Quantified Upgrade Impact
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Top 5 Bottleneck Hubs

Identified using two independent graph metrics (betweenness centrality and PageRank), with 7 of the top 10 hubs agreeing across both - this is not a single-metric artifact.

Hub	In/Out-Degree	Betweenness Rank	Network Role
IND000000ACB	46 / 49	#1 (both metrics)	Primary chokepoint
IND562132AAA	33 / 36	#2 betweenness	Secondary chokepoint
IND501359AAE	32 / 30	#2 PageRank	Tertiary chokepoint
IND712311AAA	22 / 22	#4 betweenness	Also, a top same-hub delay facility
IND421302AAG	26 / 28	#5 betweenness	Appears in top inter-hub AND same-hub lists

The Quantified Upgrade Case - and an Important Correction

Upgrading the top-3 hubs (IND000000ACB, IND562132AAA, IND501359AAE) targets a disproportionate share of network delay: these hubs handle 21.4% of trip volume but account for 35.9% of total network delay-minutes.

Important correction to a common assumption: this upgrade will NOT reduce the percentage of shipments that are late. Our analysis found the late-delivery rate (actual time exceeding OSRM estimate by >20%) is 96.1% network-wide and 96.2% for trips touching the top-3 hubs - statistically indistinguishable. Lateness against OSRM's optimistic baseline is a near-universal network property, not a hub-specific defect. The upgrade case rests on reducing delay severity and volume on the highest-traffic corridors, not on moving this rate.

Routing Correction: FTL vs. Carting

Our model-backed framework, tested across 17 corridor profiles (distance × time-of-day × origin centrality, each requiring 15+ historical trips), found a result that contradicts the default assumption that FTL is the faster option:

- Carting beats FTL on speed in 15 of 17 profiles tested - by an average of 201 minutes.
- FTL only shows a genuine time advantage in 2 profiles: Short-Haul corridors from Low-Centrality origins (average advantage: 13 minutes).

Recommendation: restrict FTL routing to the specific Short-Haul, Low-Centrality profile where it is verified faster. Default to Carting everywhere else - this is a direct, immediate routing-policy change requiring no capital investment.

Recommended Actions

- Prioritize infrastructure investment at IND000000ACB first - confirmed #1 on both independent centrality metrics, and the single largest same-hub delay contributor in the network.
- Update routing policy immediately to default to Carting except in the verified Short-Haul/Low-Centrality FTL-advantage profile - no capital cost, immediate effect.
- Separately investigate same-hub (intra-facility) handling delays as a distinct workstream from inter-hub corridor delays - 8% of all 'corridors' are same-hub pairs, representing a facility-processing problem, not a road-network problem.
- Communicate to stakeholders that hub upgrades target delay severity and volume, not the overall on-time percentage - setting this expectation correctly avoids a credibility gap when the late-rate doesn't move after the upgrade.